

Investigating a function with the derivative

Lessons 19–22 · Algebra 11, §1.7 · P1 · 7.7–7.8 · 28 problems

Solve $y' = 0$ for the stationary points, then classify each one — by the sign of y' on either side, or by y'' . Always put the x -value back into y , not into y' .

Easy 7

One stationary point, or a sign that never changes. Problem 7 is the reminder that $y' = 0$ does not have to mean a turning point.

1 Stationary point of $y = x^2 - 8x$ | $x = 4$
 $y' = 2x - 8 = 0$.

2 Classify it | minimum, at $(4, -16)$
 $y'' = 2 > 0$, so the curve is bending upwards.

3 Stationary point of $y = 6x - x^2$, and its nature | maximum at $(3, 9)$
 $y' = 6 - 2x = 0$ gives $x = 3$; $y'' = -2 < 0$.

4 Is $y = 5 - 2x$ increasing or decreasing? | decreasing everywhere
 $y' = -2$, which is negative for every x .

5 Find y'' for $y = x^4 - 2x^2$ | $12x^2 - 4$
 $y' = 4x^3 - 4x$, and differentiating again gives $12x^2 - 4$.

6 Where is $y = x^2 - 2x$ increasing? | $x > 1$
 $y' = 2x - 2 > 0$.

7 Stationary point of $y = x^3 + 1$, and its nature | $x = 0$, a stationary inflection
 $y' = 3x^2 = 0$ only at $x = 0$, but $y' \geq 0$ on both sides, so the curve flattens without turning.

Medium 7

Two stationary points, or an interval to check. In 11 and 12 one of the two is not a turning point, which the second derivative alone will not reveal.

8 Investigate $y = x^3 - 6x$ | max $(-\sqrt{2}, 4\sqrt{2})$, min $(\sqrt{2}, -4\sqrt{2})$
 $y' = 3x^2 - 6 = 0$ gives $x = \pm\sqrt{2}$. $y'' = 6x$, negative at $-\sqrt{2}$ and positive at $\sqrt{2}$. $4\sqrt{2} \approx 5.66$.

9 Investigate $y = 2x^3 - 9x^2 + 12x$ | max $(1, 5)$, min $(2, 4)$
 $y' = 6x^2 - 18x + 12 = 6(x-1)(x-2)$. $y'' = 12x - 18$ is -6 at $x = 1$ and 6 at $x = 2$. Note the maximum value here is larger than the minimum value, but neither is the greatest or least value of the function.

10 Where is $y = x^3 - 3x^2$ decreasing? | $0 < x < 2$
 $y' = 3x(x-2) < 0$ exactly between its two roots.

- 11 Stationary points of $y = x^4 - 4x^3$ | $x = 0$ and $x = 3$

$$y' = 4x^3 - 12x^2 = 4x^2(x - 3).$$

- 12 Give the nature of each | $x = 3$ minimum $(3, -27)$; $x = 0$ inflection

At $x = 0$ the factor $4x^2$ keeps the sign of y' negative on both sides, so the curve flattens and carries on down. $y''(0) = 0$, which is why the second-derivative test says nothing there.

- 13 Point of inflection of $y = x^3 - 9x^2$ | $(3, -54)$

$y'' = 6x - 18 = 0$ gives $x = 3$, and y'' changes sign there.

- 14 Greatest value of $y = x^3 - 3x$ on $[0, 2]$ | 2, at $x = 2$

Stationary at $x = 1$ with $y = -2$; the ends give $y(0) = 0$ and $y(2) = 2$. On a closed interval the ends must always be tested.

Hard 7

Unknown constants, or a sign chart with three roots. In 20 the answer is a range of k , not a number.

- 15 Investigate $y = \frac{x^2}{x^2 + 3}$ | minimum at $(0, 0)$

Quotient rule: $y' = \frac{2x(x^2 + 3) - x^2(2x)}{(x^2 + 3)^2} = \frac{6x}{(x^2 + 3)^2}$. The denominator is always positive, so the sign of y' is the sign of x : down then up, a minimum. There is no maximum — the curve rises towards $y = 1$ without reaching it.

- 16 Find a so that $y = x^3 - ax$ has stationary points at $x = \pm 2$. | $a = 12$

$y' = 3x^2 - a = 0$ at $x = 2$ gives $a = 12$, and $x = -2$ then works too because y' is even.

- 17 Find a and b so that $y = x^3 + ax^2 + bx$ has a stationary point at $(1, -2)$. | $a = 0, b = -3$

Two conditions. The point: $1 + a + b = -2$, so $a + b = -3$. Stationary: $y' = 3x^2 + 2ax + b$ gives $3 + 2a + b = 0$. Subtracting, $a = 0$ and $b = -3$. The curve is $y = x^3 - 3x$.

- 18 Show that $y = x^3 + 4x - 1$ is increasing for every x . | shown

$y' = 3x^2 + 4$. Since $3x^2 \geq 0$, the derivative is at least 4, so it is never zero and never negative. A cubic with no stationary point at all.

- 19 Find the intervals on which $y = 3x^4 - 4x^3 - 12x^2$ is increasing. | $-1 < x < 0$ and $x > 2$

$y' = 12x^3 - 12x^2 - 24x = 12x(x - 2)(x + 1)$. Three roots at $-1, 0$ and 2 split the line into four parts; testing one point in each gives the signs $-$, $+$, $-$, $+$.

- 20 $y = x^4 + kx^2$ has exactly one stationary point. Find the values of k . | $k \geq 0$

$y' = 2x(2x^2 + k)$. The factor x always gives $x = 0$. The bracket gives $x^2 = -\frac{k}{2}$, which has real roots only when $k < 0$. So one stationary point exactly when $k \geq 0$, and three when $k < 0$.

- 21 Find and classify the stationary points of $y = x^3 - 3x^2 - 9x + 5$. | max $(-1, 10)$, min $(3, -22)$

$y' = 3x^2 - 6x - 9 = 3(x - 3)(x + 1)$. $y'' = 6x - 6$ is -12 at $x = -1$ and 12 at $x = 3$.

Where the routine test breaks down, and where the conditions are given in reverse. In 25 the two stationary points are handed to you as the roots of f' .

- 22 Show that $y = x^4$ has a minimum at $x = 0$ even though $y''(0) = 0$, and explain why the second-derivative test fails. **shown**

$y' = 4x^3$, which is negative for $x < 0$ and positive for $x > 0$, so the curve falls then rises — a minimum. But $y'' = 12x^2$ is 0 at $x = 0$. The test only reports a verdict when $y'' \neq 0$; when it is zero the test is silent, and the sign of y' must be used instead. It is not evidence against a minimum.

- 23 Investigate $y = (x - 1)^2(x + 2)$ fully: stationary points, their nature, and the point of inflection.

max $(-1, 4)$, min $(1, 0)$, inflection $(0, 2)$

Expanding gives $y = x^3 - 3x^2 + 2x$, so $y' = 3x^2 - 3$ and $x = \pm 1$. $y'' = 6x$, so $x = -1$ is a maximum and $x = 1$ a minimum; $y'' = 0$ at $x = 0$. The repeated factor $(x - 1)^2$ is why the curve touches the x -axis at $x = 1$ instead of crossing it.

- 24 Find the values of k for which $y = x^3 + 3x^2 + kx$ has no stationary point. **$k > 3$**

$y' = 3x^2 + 6x + k$ must have no real root, so its discriminant $36 - 12k$ must be negative.

- 25 $f(x) = ax^3 + bx^2 + cx$ has a maximum at $x = -1$, a minimum at $x = 3$ and $f(-1) = 10$. Find a , b and c .

$a = 2$, $b = -6$, $c = -18$

$f'(x) = 3ax^2 + 2bx + c$ has roots -1 and 3 , so $f'(x) = 3a(x + 1)(x - 3) = 3a(x^2 - 2x - 3)$. Comparing coefficients, $2b = -6a$ and $c = -9a$. Then $f(-1) = -a + b - c = 5a = 10$. Check: $f' = 6(x + 1)(x - 3)$, which is positive before -1 — a maximum there, as required.

- 26 The curve $y = x^3 + px + q$ has a stationary point at $(2, -6)$. Find p and q , and the other stationary point.

$p = -12$, $q = 10$; the other is $(-2, 26)$

$y' = 3x^2 + p = 0$ at $x = 2$ gives $p = -12$. Then $y(2) = 8 - 24 + q = -6$ gives $q = 10$. Because y' is even, the other root is $x = -2$, and $y(-2) = -8 + 24 + 10 = 26$.

- 27 Find the greatest and least values of $y = x^3 - 6x^2 + 9x + 1$ on $[0, 4]$. **greatest 5, least 1**

$y' = 3(x - 1)(x - 3)$, so the stationary points are $x = 1$ and $x = 3$. The four candidates are $y(0) = 1$, $y(1) = 5$, $y(3) = 1$, $y(4) = 5$. Both extreme values are reached twice — a local maximum ties with an endpoint, and a local minimum ties with the other endpoint.

- 28 Show that if $f'(a) = 0$ and $f''(a) > 0$ then f has a local minimum at a . **shown**

$f''(a) > 0$ means f' is increasing at a . Since $f'(a) = 0$, just to the left of a the derivative f' is negative and just to the right it is positive. A function that falls then rises has a minimum at the turn. This is the whole content of the second-derivative test, and it also shows why $f''(a) = 0$ settles nothing: f' may then be increasing, decreasing or neither.